

Amir Lahlou

MSc Robotics, EPFL · Internship in industrial automation solutions from Feb 2027, six months or longer, or a 25-week EPFL master project
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EDUCATION

Master of Science in Robotics Sept 2025 – Present
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

- Minor in Systems Engineering (advanced control, stochastic processes, optimization).
- Key courses: Reinforcement Learning (**5.75/6**), Applied & Industrial Robotics (**5.75/6**), Advanced Control Systems (**5.5/6**), Robotics for Manipulation (**5.25/6**), Model Predictive Control (**5.0/6**).

Bachelor of Science in Microengineering Sept 2020 – Jul 2025
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland Final grade: **5.01/6**

EXPERIENCE

Teaching Assistant Sep 2023 – Jun 2025
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

- Supervised students in Microcontrollers & Logic Systems lab sessions (ARM Cortex-M, serial protocols, logic circuits, finite-state machines) and ran exercise sessions and grading for Electrical & Electronic Systems and Mechanical Physics.

Precision Machining Intern Summer 2024
École Technique des Métiers de Lausanne (ETML) Lausanne, Switzerland

- Operated lathes, milling and drilling machines to tight dimensional tolerances; manufactured a fully functional mechanical assembly from raw stock, combining machining and welding.

PROJECTS

Master Semester Project – Gaze-Guided Shared Autonomy Autumn 2026 – ongoing
Idiap Research Institute (HRAI group) / EPFL – Supervised by Dr. Emmanuel Senft, Prof. Jean-Marc Odobez

- Adding the operator's eye gaze to the group's joystick-based shared-autonomy framework, so the robot identifies the intended target earlier and adjusts its assistance to how sure it is; the joystick keeps the final say.

Bimanual Robotic Assembly Cell for Textile Panels – Personal Project (Simulation) 2026 – Present

- Built a physics simulation (GPU engine, Newton) of a dual-arm robotic cell: two UR5e arms manipulating deformable fabric panels, with progressive stitching of the seams simulated as well.
- Implemented arm control and a teleoperation interface to drive the assembly; alignment scored with a residual measured just before the seam is closed.

Master Semester Project – Perching Glider (5.25/6) Autumn 2025
Computer Vision Laboratory (CVLab), EPFL – Supervised by Prof. Pascal Fua

- Designed and built a 150g fixed-wing glider end-to-end as an experimental platform for perching maneuvers: low-Reynolds-number aerodynamic analysis to size the modular NACA 4412 wings, airframe design in CAD, and assembly of the physical prototype.
- Developed the onboard electronics around dual microcontrollers and programmed the wireless interface between the ground command and the aircraft (radio link driving pitch actuation).

3-Axis Cartesian Machine – Course Project, Applied & Industrial Robotics (MICRO-451) Spring 2026

- Sized a large-format Cartesian machine (1.7 m × 1.5 m travel, 3 kg payload) from the loads down to motor sizing: linear rails, GT2 belt drives, aluminium T-slot frame and crossbeam, dual-motor X axis with open-loop steppers; worst-case accuracy budget of 0.36 mm against a ±1 mm target.
- Modelled the mechanism in Fusion 360 (rails, carriages, crossbeam, belt clamps, brackets) and wrote the full design report.

Rocket Landing with Model Predictive Control – Course Project (ME-425) Spring 2026

- Designed MPC controllers in Python for a 12-state thrust-vector-controlled rocket: linear MPC on decomposed subsystems with terminal invariant sets, trajectory tracking, offset-free control under model mismatch, and nonlinear MPC on the full model.

Imitation Learning Benchmark – Reinforcement Learning Course (EE-568) Spring 2026

- Reimplemented two recent imitation-learning algorithms (CSIL, SOAR) from the original papers and benchmarked their sample efficiency across expert-demonstration budgets on Gymnasium control tasks.

ASSOCIATIONS & TEAMWORK

Xplore EPFL – Mars Rover Challenge Sept 2024 – May 2025
Project Manager & Electrical Engineer

- Led a team of 7 to design and build a competition rover, coordinating the mechanical, electrical and software workstreams.
- Designed and validated PCBs for embedded subsystems, built the power management system (distribution, protections, monitoring), tuned PID motion controllers and implemented I²C communication between sensors, actuators and microcontroller.

EPFL Racing Team – Embedded Sensor System 2022–2023

- Designed a compact embedded data-acquisition system for a Formula Student car: multi-sensor fusion, real-time data logging, and performance analytics.

TECHNICAL SKILLS

- **Embedded Systems & Software:** C, C++, Python, ARM Cortex-M, RTOS, Microcontrollers, I2C, SPI, UART, Git
- **Control & Robotics:** Model Predictive Control (MPC), LQR, PID, Nonlinear Control, Kinematics, Mobile Robotics, Signal Processing, Physics Simulation (MuJoCo, Newton), Reinforcement & Imitation Learning (PyTorch)
- **Electronics & Hardware:** PCB Design (KiCad), Analog/Digital Circuits, Sensors Integration, Power Management, A/D Conversion
- **Mechanical & Tools:** MATLAB/Simulink, CAD (Fusion 360, CATIA V5), Precision Machining (Lathes, Milling)

LANGUAGES & INTERESTS

- **Languages:** French (Native), Arabic (Native), English (Fluent).
- **Interests:** Basketball (10+ years, competitive), Origami (advanced practitioner), Philosophy (epistemology, technology).